



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

AGRICULTURAL SCIENCES P1

2023

MARKING GUIDELINES

MARKS: 150

These marking guidelines consist of 11 pages.

SECTION A**QUESTION 1**

1.1	1.1.1	B ✓✓	(10 x 2)	(20)
	1.1.2	A ✓✓		
	1.1.3	C ✓✓		
	1.1.4	C ✓✓		
	1.1.5	D ✓✓		
	1.1.6	D ✓✓		
	1.1.7	B ✓✓		
	1.1.8	D ✓✓		
	1.1.9	A ✓✓		
	1.1.10	B ✓✓		
1.2	1.2.1	B only ✓✓	(5 x 2)	(10)
	1.2.2	Both A and B ✓✓		
	1.2.3	A only ✓✓		
	1.2.4	B only ✓✓		
	1.2.5	None ✓✓		
1.3	1.3.1	Regurgitation/retro-peristalsis ✓✓	(5 x 2)	(10)
	1.3.2	Quarantine/isolation ✓✓		
	1.3.3	Urethra ✓✓		
	1.3.4	Nucleus ✓✓		
	1.3.5	Repeat-breeder syndrome ✓✓		
1.4	1.4.1	Bolus ✓	(5 x 1)	(5)
	1.4.2	Commercial ✓		
	1.4.3	Scrotum ✓		
	1.4.4	Mummification ✓		
	1.4.5	Ejaculation ✓		

TOTAL SECTION A: 45

SECTION B**QUESTION 2: ANIMAL NUTRITION****2.1 Digestive system of a farm animal**

- 2.1.1 **Name of the farm animal**
Chicken/fowl/poultry ✓ (1)
- 2.1.2 **TWO reasons visible in the diagram**
- Presence of a crop ✓
 - Presence of proventriculus/glandular stomach ✓
 - Simple stomach/monogastric ✓
 - Presence of ventriculus/gizzard/muscular stomach ✓
 - Presence of caeca/two blind guts ✓ (Any 2) (2)
- 2.1.3 **Indication of the pH**
- (a) **B - Acidic** ✓ (1)
- (b) **E - Alkaline/basic** ✓ (1)
- 2.1.4 **TWO importance of the substance/gastric juice in digestion**
- It is antiseptic and destroys bacteria/prevent rotting of the stomach content ✓
 - Changes the pH of the stomach from alkaline to acidic ✓
 - Changes disaccharides into monosaccharide ✓
 - Activates pepsinogen to form pepsin ✓
 - Pepsin changes proteins to peptones ✓ (Any 2) (2)
- 2.1.5 **Role played by part labelled C/gizzard**
It grinds food into smaller particles for easy digestion/mechanical/physical digestion ✓ (1)

2.2 Sow and its litter in a farrowing pen

- 2.2.1 **Mineral element deficient**
Iron/Fe ✓ (1)
- 2.2.2 **TWO iron deficiency symptoms**
- Anaemia ✓
 - Paleness of mucous membranes ✓
 - Listlessness/laziness/fatigue/lethargy ✓
 - Laboured/difficult breathing ✓
 - Accelerated heartbeat ✓
 - Loss of appetite ✓
 - Diarrhoea ✓
 - Reduced growth ✓ (Any 2) (2)
- 2.2.3 **A method of supplementing iron**
- Soil sods ✓
 - Injection ✓
 - Iron paste/paint/solution ✓ (Any 1) (1)

- 2.2.4 **Feed component for optimum growth of piglets**
Protein ✓ (1)

2.3 Feed trial

- 2.3.1 **The purpose of the feed trial**
To determine the digestibility of the hay/amount of oat hay digested and absorbed ✓ (1)

- 2.3.2 **Classification of the feed**
Roughage ✓ (1)

- 2.3.3 **Calculation of the digestibility co-efficiency**

$$DC = \frac{\text{Dry material intake (kg)} - \text{Dry mass of manure (kg)}}{\text{Dry material intake (kg)}} \times \frac{100}{1} \checkmark$$

$$= \frac{8 \text{ kg} - 4,5 \text{ kg}}{8 \text{ kg}} \times \frac{100}{1} \checkmark$$

$$= 43,75 \checkmark \% \checkmark$$
 (4)

- 2.3.4 **Naming the substances**
 (a) Molasses ✓ (1)
 (b) Urea/biuret ✓ (1)

2.4 Nutritive ratio

- 2.4.1 **Calculation**
 (a) **% of non-nitrogen substances**

$$= 80\% - 8\% \checkmark$$

$$= 72\% \checkmark$$
 (2)

- (b) **Nutritive ratio**

$$NR = 1 : \frac{\%TDN - \%DP}{\%DP} \checkmark$$

$$1 : \frac{80\% - 8\%}{8\%} \checkmark$$

$$1 : 9 \checkmark$$

OR

$$NR = 1 : \frac{\% \text{ digestible non-nitrogen substances}}{\% \text{ digestible protein}} \checkmark$$

$$1 : \frac{72\%}{8\%} \checkmark$$

$$1 : 9 \checkmark$$
 (3)

- 2.4.2 **TWO components making up non-nitrogen content in a feed**
 • Digestible fat/lipids ✓
 • Digestible carbohydrates ✓
 • Vitamins ✓
 • Minerals ✓ (Any 2) (2)

2.5 Feed flow programme**2.5.1 Calculation of the total amount of feed required**

Feed required = number of animals x feed/animal/day x number of days

= 150 animals x 5 kg x 30 days ✓

= $\frac{22\ 500}{1\ 000}$ ✓

= 22,5 tons ✓

(3)

2.5.2 Quantity of cattle feed during month 3

Feed will be enough ✓

(1)

2.5.3 Reason

Feed required is 22 500 kg and feed available is 30 000 kg/there is a surplus of 7 500 kg ✓

(1)

2.5.4 The month with the least shortage of feed

Month 6 ✓

(1)

2.5.5 ONE cost effective strategy to address the shortage of feed

- Storage feed during months where there is an excess ✓
- Stock reduction/culling ✓
- Controlled calving/change the breeding season ✓
- Planting of seasonal fodder crops ✓

(Any 1)

(1)

[35]**QUESTION 3 : ANIMAL PRODUCTION, PROTECTION AND CONTROL****3.1 Production output and cost distribution for two feedlots****3.1.1 The feedlot which operates at the highest cost**

Feedlot 2 ✓

(1)

3.1.2 The feedlot which operates in the most cost-efficient way

Feedlot 1 ✓

(1)

3.1.3 Explanation of the answer in QUESTION 3.1.2

- The total cost was the lowest/R780 compared to R810 ✓
the output was the highest/R1 720 compared to R1 680 ✓
- A greater output ✓ for a lower cost ✓

(Any 1)

(2)

3.2 How the structures help the farm animals to survive adverse environmental conditions

(a) **Shelter** - Has sides for protection against cold winds/reduce the wind chill/the enclosed area keeps heat within/insulation ✓

(1)

(b) **Insulation material** - Heat can be retained/protection against cold/heat for a longer period of time/cooling effect ✓

(1)

(c) **Roofing** - For protection against rain/cold/direct sunlight ✓

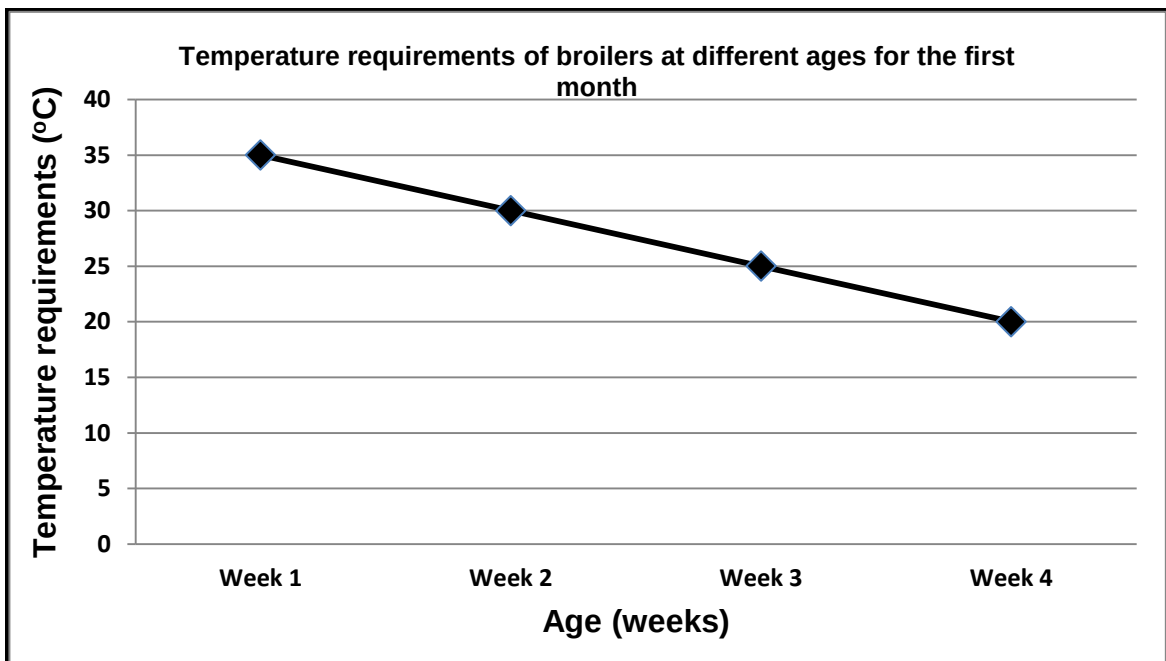
(1)

3.3 Temperature requirements of broilers at different ages

3.3.1 **The temperature requirements at three weeks**
25°C ✓ (1)

3.3.2 **The trend of temperature requirements over a period of 7 weeks**
Temperature requirements decline with increased age ✓ until it stabilizes from week 5 to 7 ✓ (2)

3.3.3 **Line graph showing the temperature requirements of broilers at different ages for the first month**

**CRITERIA/RUBRIC/MARKING GUIDELINES**

- Correct heading ✓
- X-axis: correct calibrations and labelling (Age) ✓
- Y-axis: correct calibrations and labelling (Temperature requirements) ✓
- Correct units (°C and weeks) ✓
- Line graph ✓
- Accuracy (80% + correctly plotted) ✓ (6)

3.4 Handling facility

3.4.1 **Identification of the handling facility**
Holding pen/paddock ✓ (1)

3.4.2 **TWO reasons for restraining farm animals in a crush**

- To ensure safety while working with large animals/no harm to the handlers ✓
- To be able to work with animals while they are stable ✓
- To perform specialised practices on animals/AI/dehorning/castration/tattooing/branding/medication/physical examinations ✓
- Time and labour efficient ✓ (Any 2) (2)

3.4.3 TWO basic guidelines when handling cattle

- Keep safety as the main principle in mind ✓
- Cattle should be kept as calm as possible ✓
- Use the correct handling equipment/facilities ✓
- No carrying of sticks/beating/throwing stones ✓
- No shouting/whistling/wild gestures ✓
- Move around slowly/no running around ✓
- Keep animals of the same size/age/sex together ✓
- Separate sick/old/pregnant animals from healthy animals ✓
- Limit the number of people in a facility ✓
- Do not approach animals from behind ✓
- Announce your presence through touch to the animal's front or side ✓
- Let cattle in and out in the same manner/use routine ✓ (Any 2) (2)

3.5 Animal diseases**3.5.1 Completing the table on animal diseases**

- A** - Virus ✓ (1)
- B** - Mastitis ✓ (1)
- C** - Poultry/cattle/pigs/sheep ✓ (1)
- D** - Fleece contains hard lumps/crusts/scabs on the ears/lips/face/
shanks/scrotum/sores on the skin/loss of wool ✓ (1)

3.5.2 TWO preventative measures for controlling Newcastle disease

- Timely diagnosis and vaccination against diseases ✓
- Outbreaks must be detected quickly ✓
- Good husbandry must be practiced/housing/nutrition/
management ✓
- Movement of animals should be controlled ✓
- Proper hygiene/sanitation/clean ✓
- Use breeds resistant to diseases ✓
- Quarantine/isolation ✓
- Proper disposal of carcasses ✓ (Any 2) (2)

3.5.3 TWO financial implications of animal diseases

- Decrease in production/poor quality ✓
- Stock losses/death ✓
- Decrease in income/profit ✓
- Banning of exports/international trade decreases ✓
- Have negative impact on food security ✓
- High cost to control/treatment ✓ (Any 2) (2)

3.6 Letters representing stages in the life cycle of the parasite

- (a)** B ✓ (1)
- (b)** D ✓ (1)
- (c)** E ✓ (1)
- (d)** C ✓ (1)

3.7 TWO symptoms of urea poisoning in farm animals

- Nervous symptoms/lack of balance/incoordination ✓
- Excessive salivation ✓
- Frequent defecation and urination ✓
- Struggling violently/bellowing ✓
- Bloating ✓
- Tetany/muscular pain ✓
- Breathing difficulty ✓
- Rapid death ✓

(Any 2) (2)
[35]

QUESTION 4: ANIMAL REPRODUCTION**4.1 The reproductive systems of farm animals****4.1.1 The letter representing the part in the diagrams**

- (a) D ✓ (1)
 (b) G ✓ (1)
 (c) B ✓ (1)

4.1.2 A membrane responsible for implantation

Endometrium ✓ (1)

4.1.3 The part performing the same function as the testis

C ✓ (1)

4.1.4 TWO congenital defects of the testes

- Cryptorchidism ✓
 - Hypoplasia ✓
- (2)

4.2 Pie chart**4.2.1 Duration of the oestrus cycle in cows**

21 days ✓ (1)

4.2.2 Oestrus stages

- (a) B - Di-oestrus ✓ (1)
 (b) D - Oestrus ✓ (1)

4.2.3 The letter representing the stage of oestrus cycle

C ✓ (1)

4.2.4 TWO practical methods to identify a dairy cow in heat

- Heat mount detector ✓
 - Tail chalking/tail head marker ✓
 - Pedometer ✓
 - Chin-ball marker ✓
 - Heat observation ✓
 - Androgenised females ✓
- (Any 2) (2)

4.2.5 What happens to the corpus luteum if the cow becomes pregnant

Corpus luteum persists and continue to secrete progesterone ✓ (1)

4.3 Mating behaviour**4.3.1 TWO factors regulating mating behaviour**

- Hormonal influences ✓
- Social interaction ✓
- Senses/sight/smell ✓
- Environmental factors ✓
- Physiological factors ✓
- Previous experience ✓
- Health ✓
- Genetic factors ✓
- Libido ✓

(Any 2) (2)

4.3.2 Substance in cow's urine that stimulates libido in bulls

Pheromones ✓

(1)

4.4 Reproductive techniques**4.4.1 Identification of a reproductive technique****(a)** Cloning ✓

(1)

(b) Synchronization of oestrus/embryo transfer ✓

(1)

4.4.2 Definition of embryo transfer

The transfer of embryo from a genetically superior female animal (donor) ✓ to the uterus of genetically inferior female animal (recipient) ✓

(2)

4.4.3 TWO advantages of Artificial insemination

- Reduces the exchange of sexually transmitted diseases ✓
- Superior male animal can fertilize more female animals ✓
- Semen from males in other countries can be used ✓
- Quick and economical way to improve the herd ✓
- Valuable tool in assisting with progeny testing ✓
- Semen can be used long after bull's death ✓
- Possible where mating is impossible ✓
- No need to buy, keep and maintain expensive bulls ✓
- Inferior bulls can be detected at an early stage and avoided ✓
- AI increases the reproductive and conception rate ✓

(Any 2) (2)

4.4.4 Reproductive stage following immediately after successful insemination

Fertilisation ✓

(1)

4.5 Parturition**4.5.1 Scientific term for difficult birth**

Dystocia ✓

(1)

4.5.2 TWO causes of dystocia

- Deviation of the head ✓
- Flexion of elbow ✓
- Retention of one or both forelegs ✓
- Hydrocephalus ✓
- Congenital defects/abnormalities/malformed foetus ✓
- Twinning/multiplets ✓
- Posterior/incorrect presentation ✓
- Age of the animal ✓
- Large foetus ✓
- Dead foetus ✓
- Torsion of the uterus
- Uterine inertia/weak contractions/labour ✓
- Prolonged gestation period ✓
- Size of the pelvic area ✓
- Poor body condition ✓
- Incomplete cervical dilation ✓
- Vaginal tear/injuries ✓
- Diseases ✓

(Any 2) (2)

4.5.3 Hormone responsible for the relaxation of the cow's muscles

Relaxin ✓

(1)

4.5.4 THREE noticeable behavioural changes in a cow which is about to give birth

- Isolation/nesting behaviour ✓
- Stops eating/lack of appetite ✓
- Making bellowing noises ✓
- Restlessness/signs of discomfort because of pain ✓
- Urinates and defecates often ✓

(Any 3) (3)

4.6 Milk production in cows**4.6.1 The hormones responsible for****(a) Milk production - Prolactin ✓**

(1)

(b) Milk release - Oxytocin ✓

(1)

4.6.2 TWO stimuli enhancing the release of milk

- Touching/washing/massaging of the udder ✓
- Sound of the milking machine ✓
- Presence/sight of the calf/suckling of the calf ✓
- Presence of the milker ✓

(Any 2) (2)
[35]

TOTAL SECTION B: 105
GRAND TOTAL: 150